CSC3105

Data Analytics Sample Quiz 1

Sample Quiz 1 - Trimester 2, AY2025/26

xx, xxth Feb 2026

x.xxam – x.xxam

(2 hours)

Instructions to students:

1. This Quiz comprises **FOUR(4)** Questions printed on **Fifteen (15)** pages including the cover page.
2. You must answer **ALL** the questions.
3. All answers must be written in this answer sheet.

At the end of the examination,

Please write your **Student ID** on each answer book used.

Failure to do so will mean that your work will not be identified.

The University reserves the right not to mark your script if you fail to follow these instructions.

Answer all the questions (Total 100 marks)

Question 1 (20 marks)

Google search engine deal with more than 3.5 billion searches per day. The main reason for quick query response is that Google uses Term Frequency-Inverse Document Frequency (TF-IDF) approach behind the scene to match each query in millions of documents and websites using proximity measures. It will match those websites and documents with the query and score them according to the proximity measure used. Hence, it will rank and return those that are highly matched in descending order accordingly.

Assume that Google has computed millions of website and documents with the following TF-IDF scores for the following websites that contains following wordings as followed.

Website 1 : Data Analytics is not about statistical calculus mathematics and algorithms but physical insight and visualization on data with decision making
Website 2: Data Analytics seems daunting with mathematics and algorithms tools but is fun decision tools
Website 3: Data Analytics is linked to data decision making tools with data visualization

For these 3 websites, the TF-IDF scores for the wordings "Data" , "Analytics" are as follows:

	Website 1	Website 2	Website 3
Data	0.0307	0.0219	0.0921
Analytics	0.05	0.0714	0.1

For the user query on the word Data Analytics , the TF-IDF score is as follows:

	TF-IDF
Data	0.1534
Analytics	0.5

Using the cosine similarity proximity measure on the TF-IDF of the user query and the three websites, find the website that the Google will rank the highest and return the response back to user based on cosine similarity.

Repeat the approach with the Euclidean distance proximity measure on the TF-IDF of the user query and the three websites to find the website that the Google will rank the highest and return the response back to user.

Hint: $\mathrm{d}(\mathbf{x},\mathbf{y}) = \|\mathbf{x}-\mathbf{y}\|$

$\cos(\mathbf{x},\mathbf{y}) = \frac{<\mathbf{x},\mathbf{y}>}{\|\mathbf{x}\|\|\mathbf{y}\|}$

Solution :

Question 2 (30 marks)

Given a data matrix $\mathbf{x} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ where column represents the two features namely number of Mac user and number of Window users respectively. The row represents the two data record. Derive out the PC components of this data matrix x. Obtain the dimension reduction using just 1 PC component.

Hint: $\text{cov}(\mathbf{x}) = \frac{1}{N-1}(\mathbf{x} - \boldsymbol{\mu})^T(\mathbf{x} - \boldsymbol{\mu})$

<u>Solution :</u>

Question 3 (30 marks)

Given the table below, find out which features should be the best split to be the main first node.

Name	Can Fly	Live In Water	Class
Human	No	No	Mammal
Python	No	No	Non- Mammal
Salmon	No	Yes	Non- Mammal
Whale	No	Yes	Mammal
Komodo	No	No	Non- Mammal
Bat	Yes	No	Mammal
Pigeon	Yes	No	Non- Mammal
Cat	No	No	Mammal
Leopard Shark	No	Yes	Non- Mammal
Porcupine	No	No	Mammal
Eel	No	Yes	Non- Mammal
Gila Monster	No	No	Non- Mammal
Platypus	No	No	Mammal
Owl	Yes	No	Non- Mammal
Dolphin	No	Yes	Mammal
Eagle	Yes	No	Non- Mammal

Hint : $Entropy = -\sum_{i=1}^{N} p_i \, log_2(pi)$ where N is the number of classes

<u>Solution :</u>

Question 4 (20 marks)

Suppose you have the following instance data where x and y are the 2 input features and the class label L

Instance	x	y	Class label L
1	-1	1	-
2	0	1	+
3	0	2	-
4	1	-1	-
5	1	0	+
6	1	2	+
7	2	2	-
8	2	3	+

a) Draw the scatterplot in 2D space

<u>Solution :</u>

b) Suppose you want to predict the class of a new data point at $x = 1, y = 1$ using Euclidean distance in KNN where $K = 3$. In which class this data point belong to. Prove your working

Solution :

c) If we use KNN with $k = 7$, which of the class does the data point $x = 1, y = 1$ belong to. Prove your working

Solution :

---END OF PAPER---